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Software Engineering I: Practice Web Quiz

Instructions: This quiz is open-notes, closed-computer.

1. For each of the following languages, circle whether their code is executed (run) server-side, client-side, or both:

HTML server-side client-side

CSS server-side client-side

Jinja server-side client-side

Python server-side client-side

MySQL server-side client-side

2. The Flask Python code `request.values.get("something")` will NEVER return this data type:

a. NoneType

b. string

c. int

"something"
Quotes indicate String

Use the following code to answer questions 3-5:

```
<form method="GET" action="page2">  
  <input type="radio" name="results" value="value1">  
  <input type="radio" name="results" value="value2">  
  <input type="number" name="quantity">  
  <input type="submit" value="next">  
</form>
```

3. In the form above, hitting "submit" will perform a GET request to which base URL?

- a. localhost:5000/results
- b. localhost:5000/page2
- c. localhost:5000/next

action = "page2"

4. After the form above is submitted, the Python code

`request.values.get("abcde")` will return which value?

- a. None
- b. False
- c. ""
- d. [it will throw an error]

There is no part of the form named "abcde"

5. If the user enters the number 10 into the number input, which of the following is a possible resulting URL?

- a. localhost:5000/results?quantity=10
- b. localhost:5000/page2?quantity=10
- c. localhost:5000/page2?results=value1&number=10
- d. localhost:5000/page2?results=value2&q=10

The radio buttons are not selected.

6. Consider the following code:

base.html.j2

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>
5       {% block title %}{% endblock %}
6     </title>
7   </head>
8   <body>
9     <h1>My Website</h1>
10    {% block content %}{% endblock %}
11    <p>Copyright 2025</p>
12  </body>
13 </html>
```

about.html.j2

```
1 {% extends "base.html.j2" %}
2
3 {% block title %}About Us{% endblock %}
4
5 {% block contents %}
6   <h2>My Awesome Name<h2>
7   {% if name %}
8     <p class="name">{{ name }}</p>
9   {% endfor %}
10  <p>We are passionate about coding!</p>
11 {% endblock %}
```

When you run the code, you notice the following errors. Explain how you would fix each error. You may find it helpful to reference the line numbers.

a. Jinja2.exceptions.TemplateSyntaxError: Encountered unknown tag 'endfor'.

Line 9 in "about" has a `{% endfor %}` but there's no corresponding for loop. It should be `endif`

b. Nothing contained in `{% block contents %}`, such as "My Awesome Name", shows on the page.

The block is called "content" on line 10 of "base" and "contents" on line 5 of "about"

c. HTML validation error; "Unclosed h2 tag"

Line 6's closing tag should be `</h2>` and not `<h2>`

d. Could you change `<p class="name">` to `<p id="name">`? If not, why not? If so, what would have to change in your CSS file to accompany this change?

Yes you can as it only shows up once, in your CSS you'd have to change from `.name` to `#name`

7. Assume you have a database to represent a list of people and their favorite dining spots. The tables are set up as follows.

practicequiz_people		
id	first_name	last_name
1	Amanda	O'Neal
2	Michele	Lombardi
3	Michele	Clark

practicequiz_restaurants			
id	restaurant_name	restaurant_type	person_id
1	The Hot Spot	Lunch	2
2	Eat at Joe's	Dinner	3
3	The Hot Spot	Lunch	1
4	Restaurant Cafe	Breakfast	2

- (a) Fill in the missing code in the Python snippet below to query the tables above.
- The query should retrieve the `restaurant_name` of the favorite restaurants of anyone whose first name is `firstname` and last name is `lastname`.
 - The query should sanitize inputs so that no one can use SQL injection to hack it.
 - For full credit, you should do this using a single query with a JOIN.

```
mysql = MySQL(app) #Assume this is set up correctly
```

```
@app.route('/results', methods=['POST'])
```

```
def results():
```

```
    firstname = request.values.get("firstname")
```

```
    lastname = request.values.get("lastname")
```

```
    #Your code starts here
```

```
    query = """
```

```
        SELECT restaurant_name
```

```
        FROM practicequiz_restaurants
```

```
        JOIN practicequiz_people
```

```
        ON practicequiz_restaurants.person_id =
```

```
        practicequiz_people
```

```
        WHERE first_name = %s AND last_name = %s
```

```
    """
```

```
    cursor = mysql.connection.cursor(dictionary=True)
```

```
    cursor.execute(query, (firstname, lastname))
```

```
    data = cursor.fetchall()
```

```
    #Your code ends here
```

```
    return render_template('results.html.j2', rows=data)
```

fill in
first & last
name
AKA
tuple
values

multi-line
string

render template
OKS for
Ver None? data

(b) Write the contents of the <body> for a valid "results.html.j2" file. This page should:

- Work correctly with the "app.py" in part 7(a). You may assume part 7(a) is done correctly, even if you know you made mistakes in your actual answer.
- If there were no results for the query (meaning no one in the database had the entered name), show "Zero results" in a paragraph (<p>)
- If there were one or more results in the query, list the restaurant names from the query in a list. Example output for name "Amanda O'Neal":

```
<ul>
  <li>The Hot Spot</li>
</ul>
```

- No CSS required.
- Do NOT assume there is a base.html.j2 file. Do NOT use Jinja 'extends'.

results.html.j2:

```
<body>
  <!-- your code here -->
  {% if rows %}
    <ul>
      {% for row in rows %}
        <li>{{ row['restaurant_name'] }}</li>
      {% endfor %}
    </ul>
  {% else %}
    <p>Zero results</p>
  {% endif %}
</body>
```


8. Imagine we wanted to add client-side and server-side validation to the code from #7 to ensure that we never query the database with a null or empty firstname or lastname.

Which two languages would we use to do this validation?

- a. HTML for client-side; Jinja for server-side
- b. HTML for client-side; Python for server-side
- c. Jinja for client-side; Python for server-side
- d. HTML for client-side; CSS for server-side

9. Imagine you execute the query "SELECT * FROM practicequiz_people;", and save the results into a Python variable called rows. Which of the following is a valid piece of data in rows?

- a. rows['first_name']
- b. rows[0]['first_name']
- c. rows['first_name'][0]
- d. rows[0]['Amanda']

mysql.execute() returns a
list of dictionaries. You need
to index the list with [0], then
access the dictionary key
↓
['first_name']

SQL

NOTES

SELECT:

SELECT column1, column2 FROM table_name;

 or *

→ All rows matching query, with possibly restricted columns

WHERE:

SELECT * FROM table WHERE column = value;

column name
↓

WHERE col1 = val1 AND col2 = val2;

→ filtered rows

→ filter rows by condition

↑ ↑
conjunction OR

JOIN:

SELECT * FROM table1 JOIN table2 ON

→ start with table1

→ connect to table2

→ the link

→ filter results with where

[table1.var1 = table2.var2 WHERE
table1.var1 = thing]

* JOIN Basically creates a
new table with matching
data related.

GET vs POST:

request.values

request.args

request.form

either

GET (url)

POST

anytime
↓

@app.route('/results', methods=

default is GET only

['GET'])